

LEXINGTON BLUE GRASS, KY, USA

WMO: 724220

Lat: 38.041N

Lon: 84.606W

Elev: 980

StdP: 14.18

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93820

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.7	(c) 13.3	(d) -1.7	(e) 5.2	(f) 11.3	(g) 3.0	(h) 6.7	(i) 15.7	(j) 25.2	(k) 50.9	(l) 22.8	(m) 46.2	(n) 6.4	(o) 310	(p) 0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	91.5	73.8	89.4	73.4	87.3	72.6	77.3	87.2	76.0	85.3	74.9	83.5	7.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.2	132.6	82.8	73.1	127.4	81.2	72.1	122.9	79.9	41.2	87.4	40.0	85.7	38.9	83.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.9	(b) 18.6	(c) 16.6	(e) 0.4	(f) 95.2	(g) 6.8	(h) 3.4	(i) -4.5	(j) 97.7	(k) -8.5	(l) 99.7	(m) -12.4	(n) 101.6	(o) -17.3	(p) 104.1		
(4) 20.9	(5) 18.6	(6) 16.6	(7) 0.4	(8) 95.2	(9) 6.8	(10) 3.4	(11) -4.5	(12) 97.7	(13) -8.5	(14) 99.7	(15) -12.4	(16) 101.6	(17) -17.3	(18) 104.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.2	33.1	37.2	45.5	56.2	65.2	73.1	76.2	75.6	69.3	57.7	46.0	37.7		
	DBStd	17.64	12.49	11.70	10.97	9.30	7.82	5.46	4.46	4.67	7.11	8.93	9.60	10.62		
	HDD50	1809	541	381	219	45	3	0	0	0	29	188	403			
	HDD65	4483	989	778	608	285	99	8	0	1	39	258	571	847		
	CDD50	4057	17	23	80	231	473	694	811	792	578	269	68	21		
	CDD65	1261	0	0	4	22	105	253	347	329	167	33	1	0		
	CDH74	10561	0	2	21	161	774	2075	3018	2821	1427	256	6	0		
	CDH80	3634	0	0	0	15	176	695	1134	1052	509	53	0	0		
(14) Wind	WSAvg	7.8	9.5	9.2	9.3	9.2	7.5	6.7	5.9	5.7	6.2	7.2	8.3	8.7		
(15) Precipitation	PrecAvg	47.0	3.1	3.3	4.4	4.1	5.0	4.4	5.0	3.9	3.4	3.0	3.3	4.1		
	PrecMax	72.1	6.4	10.2	13.8	12.7	10.9	11.7	10.3	11.2	10.4	7.4	7.7	10.2		
	PrecMin	31.9	0.4	0.7	1.0	1.1	1.2	0.6	1.3	0.2	0.4	0.3	0.4	0.6		
	PrecStd	9.0	1.5	2.0	2.4	2.3	2.4	2.3	2.0	2.1	2.3	1.9	1.6	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.6	70.4	77.0	82.1	87.4	91.9	94.5	94.6	93.6	85.1	73.9	66.7		
		MCWB	59.1	60.0	62.4	64.7	70.6	73.6	76.3	74.2	70.7	68.7	62.8	59.3		
	2%	DB	61.4	64.3	71.7	78.8	84.3	89.1	91.2	91.1	89.5	80.4	69.0	61.8		
		MCWB	56.8	55.4	59.0	63.7	69.1	73.5	75.1	73.8	70.5	66.4	59.0	57.0		
	5%	DB	57.4	59.8	66.9	75.1	81.9	86.8	88.8	88.6	86.0	76.4	65.0	57.9		
		MCWB	53.7	53.1	56.4	61.8	68.5	72.4	74.3	73.3	70.0	64.2	57.4	53.7		
	10%	DB	52.4	55.4	62.6	71.5	78.8	84.4	86.4	86.0	82.2	72.3	61.3	54.0		
		MCWB	47.7	48.9	53.9	59.9	67.5	71.5	73.3	72.6	68.4	63.0	54.4	50.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.9	62.1	64.3	68.9	73.6	77.6	79.4	78.5	75.5	72.7	65.1	61.7		
		MCDB	63.8	68.2	74.0	78.5	82.5	87.2	89.1	88.6	86.1	81.1	71.3	65.5		
	2%	WB	57.8	58.1	61.1	66.1	71.6	75.9	77.6	76.9	73.6	69.5	62.2	57.7		
		MCDB	60.8	62.6	68.9	74.8	81.3	85.1	87.9	86.6	83.5	76.0	67.2	60.4		
	5%	WB	54.1	54.2	58.4	63.8	70.2	74.6	76.4	75.5	72.1	66.7	58.8	54.5		
		MCDB	57.1	58.6	65.0	71.7	78.9	83.6	85.7	84.7	81.1	73.5	63.3	57.4		
	10%	WB	48.3	49.5	55.2	61.6	68.6	73.2	75.1	74.3	70.7	64.2	55.3	50.3		
		MCDB	51.8	54.9	61.8	69.3	76.3	81.8	83.2	82.6	78.7	70.7	60.3	53.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	16.6	18.5	20.1	18.5	18.5	18.0	18.8	20.0	19.4	17.4	14.6		
		MCDBR	17.9	22.0	22.3	23.3	20.9	21.0	20.8	21.5	24.1	22.9	21.0	18.2		
		MCWBR	14.4	16.7	13.8	13.3	10.1	9.2	8.3	8.3	9.6	11.5	14.2	15.0		
		MCDBR	16.6	20.5	19.2	19.3	18.0	18.3	18.1	17.5	18.9	18.7	17.9	17.0		
	5% WB	MCWBR	14.2	17.4	13.2	12.7	9.9	9.1	8.4	8.0	8.9	11.3	14.3	15.2		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.320	0.349	0.390	0.436	0.447	0.472	0.466	0.408	0.353	0.320	0.299		
	taud		2.464	2.408	2.388	2.273	2.189	2.196	2.139	2.144	2.278	2.428	2.476	2.492		
	Ebn at Noon		278	288	289	282	270	266	258	256	266	272	271	273		
	Edh at Noon		26	31	35	42	46	46	48	47	39	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		610	826	1174	1566	1754	1959	1850	1740	1437	1072	743	528		
	RadStd		66	118	106	115	164	102	102	98	152	123	82	53		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+272	+166
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+135

Nomenclature: See separate page